Akoizo Original Practice - Form 1 - Answer key

Division B | Boomilever | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) A structure weighs 8.00 g and holds 15,000 g without the bonus. What is its score?

A. 1,875

B. 120

C. 15,000

D. 1,500

Correct: A

2. (2 points) How may participants keep the bucket from swinging during loading?

A. By resting it on a stool

B. By pushing the structure

C. Only with the tips of the stabilizing sticks

D. By holding the bucket handle

Correct: C

3. (2 points) Doubling a strut's unbraced length changes its Euler buckling load by a factor of:

A. 2

B. 4

C. 1/4

D. 1/2

Correct: C

4. (4 points) Calculate the moment of a 147 N load hung 0.44 m from the wall.

answer: 147 N x 0.44 m about 64.7 N·m. (4 points).

5. (2 points) Why change only one variable between test builds?

Unit: Testing, failure analysis, and iteration.
Worked-case data: Your log shows: 9.2 g held 15 kg; 8.4 g held 15 kg; 7.9 g failed at 13.1 kg (strut buckled). What should the competition build be?
Separate investigation observations:
Builds 1-3: 8.1 g, 8.3 g, 8.2 g: all held 15 kg.
Builds 4-5: 7.4 g, 7.5 g: failed at 12 kg and 14 kg, strut buckled.
Build 6: 7.6 g with a stiffer, denser strut: held 15 kg.

A. So any change in result can be traced to that variable

B. It is required by the rules

C. It saves glue

D. It makes the structure lighter

Correct: A

6. (4 points) Why are adhesive tapes and glue-soaked wood both banned, and what does that mean for design?

Unit: Joints, adhesives, and the bolt hook.
Worked-case data: A chord joint must carry 300 N. Your glue holds about 2 N per mm^2 of long-grain overlap. How much bonded area do you need, with a safety factor of 1.5?
Separate investigation observations:
Failure 1: Hook crushed into an oval around the bolt at 10 kg.
Failure 2: Strut-to-chord butt joint split at 13 kg.
Fix trial: Basswood bearing pad (+0.15 g) and two gussets (+0.10 g): held 15 kg.

answer: Adhesive may only join pieces; tape is excluded, and glue cannot be used to reinforce members. Strength must come from wood choice and joint design. (4 points).

7. (4 points) What does a pyramid layout gain over a single flat triangle?

Unit: Geometry: contact depth, strut angle, and the bonus line.
Worked-case data: Compare strut forces for a full 15 kg (147 N) at 42 cm reach, with contact at 20 cm versus 15 cm.
Separate investigation observations:
Design X: Reach 45 cm, contact 19 cm: strut needed 2 braces.
Design Y: Reach 41 cm, contact 19 cm: strut needed 1 brace.
Design Z: Reach 41 cm, contact 14 cm: strut needed 3 braces.

answer: Two spread-out chords and struts resist twisting and sideways tipping of the load, which a single plane triangle cannot. (4 points).

8. (2 points) Define “Load Scored” in the context of Boomilever.

answer: Load Supported plus the 5,000 g bonus when the bonus rules are met. (2 points).

9. (4 points) List three situations that end loading.

Unit: The test apparatus, setup, and loading procedure.
Worked-case data: Plan the six-minute window for a team that has practiced setup.
Separate investigation observations:
Run 1: Setup took 5 min; fast pour; failed at 9 kg with violent swing.
Run 2: Setup took 2 min; slow pour with stick tips; held 15 kg.
Inspection: Both structures were identical builds from the same wood batch.

answer: The structure fails; the load is supported by something other than the structure; the structure receives vertical support besides the bolt; or time expires. (4 points).

10. (4 points) Can a team glue three balsa strips into one thicker strut? Explain.

answer: Yes; lamination by the team with adhesive is allowed. Only commercially laminated wood such as plywood is prohibited. (4 points).

11. (2 points) Define “Moment” in the context of Boomilever.

answer: Force times perpendicular distance; the turning effect about a point. (2 points).

12. (6 points) Solve this case and explain your reasoning: Your log shows: 9.2 g held 15 kg; 8.4 g held 15 kg; 7.9 g failed at 13.1 kg (strut buckled). What should the competition build be?

reasoning: Scores: 1,630; 1,786; 1,658.
The 7.9 g build failed by buckling, so the strut is at its limit near that mass.
Build two or three more at about 8.4 g to confirm they hold reliably. (3 points).

conclusion: Compete with the 8.4 g design once it repeats; push lighter only with a stiffer strut, not by thinning it. (3 points).

13. (2 points) Why did the fix work?

Supplied synthetic observations:
Failure 1: Hook crushed into an oval around the bolt at 10 kg.
Failure 2: Strut-to-chord butt joint split at 13 kg.
Fix trial: Basswood bearing pad (+0.15 g) and two gussets (+0.10 g): held 15 kg.

A. It lowered bearing stress at the bolt and increased joint glue area

B. It made the strut longer

C. It moved the chain farther from the wall

Correct: A

14. (2 points) Define “Strut angle” in the context of Boomilever.

answer: Angle of the compression member from horizontal; steeper means less force. (2 points).

15. (2 points) Define “Contact depth line” in the context of Boomilever.

answer: A horizontal wall line (10, 15, 20 cm below the bolt) limiting where the structure may touch. (2 points).

16. (2 points) Define “Contact width lines” in the context of Boomilever.

answer: Vertical lines 4.0 cm either side of the bolt center. (2 points).

17. (2 points) Define “Butt joint” in the context of Boomilever.

answer: Members meeting end-to-face, with little glue area. (2 points).

18. (2 points) Define “Unbraced length” in the context of Boomilever.

answer: Distance between points that stop a member from bowing sideways. (2 points).